

Project Design Phase Solution Architecture

Date	10 July 2026
Team ID	SWTID-2026-6119
Project Name	Credit Card Approval Prediction System
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture is a complex process - with many sub-processes - that bridges the gap between business problems and technology solutions. Its goals are to:

- Find the best tech solution to solve existing business problems.
- Describe the structure, characteristics, behavior, and other aspects of the software to project stakeholders.
- Define features, development phases, and solution requirements.
- Provide specifications according to which the solution is defined, managed, and delivered.

Solution Architecture Diagram: Credit Card Approval Prediction System

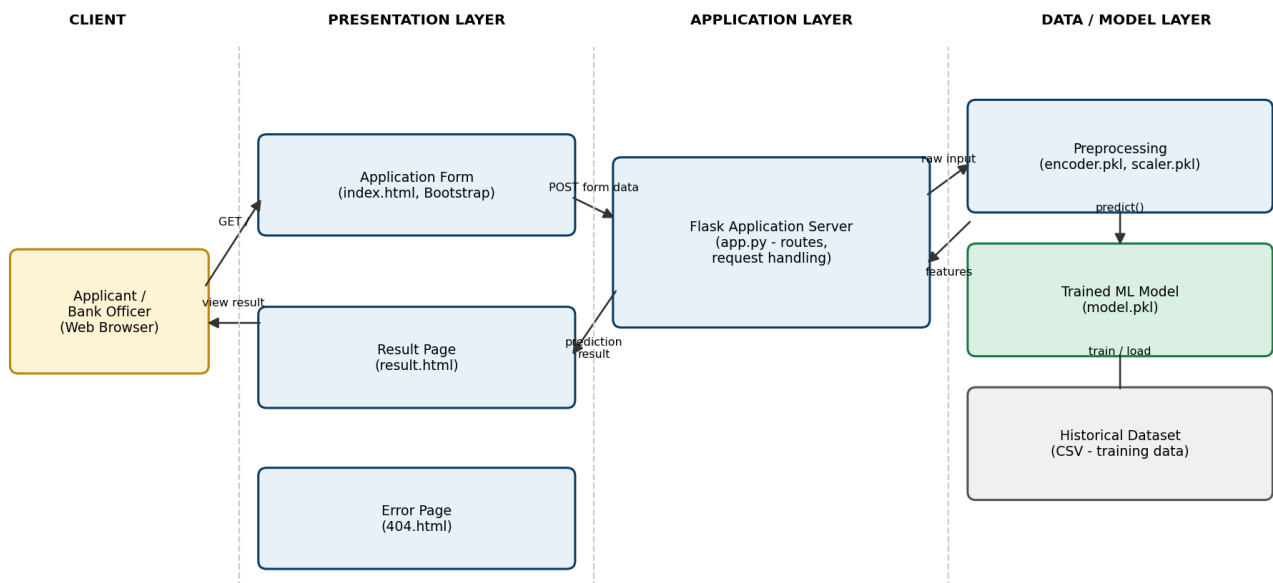


Figure 1: Architecture and data flow of the Credit Card Approval Prediction System, from applicant request through the Flask application, preprocessing pipeline, and trained ML model, back to the result page.

Layer-wise Description:

Layer	Description
Client	Applicants and bank officers access the system directly via a standard web browser, with no need to install a separate application.
Presentation Layer	Flask renders HTML/Bootstrap templates for the application form, the prediction results page, and a custom error page for invalid routes.
Application Layer	The Flask server (app.py) processes incoming form submissions, manages request routing, and links the presentation layer with the model layer.

Layer	Description
Data / Model Layer	The preprocessing objects (encoder, scaler) convert raw form inputs into features the model can use; the trained model (model.pkl) produces the Approved/Rejected result; the historical dataset supports offline training and evaluation.